Ma 6a Midterm Om Sanan

1.

We want to find all integers x s.t.

$$2x \equiv 5 \pmod 7, \quad 3x \equiv 4 \pmod 7.$$

First, solving $2x \equiv 5 \pmod 7$:

Find the inverse:

$$2 \cdot 4 = 8 \equiv 1 \pmod 7 \longrightarrow 2^{-1} \equiv 4 \pmod 7.$$

Therefore:

$$x \equiv 5 \cdot 4 = 20 \equiv 6 \pmod 7.$$
$$x \equiv 6 \pmod 7.$$

Then, solving $3x \equiv 4 \pmod 8$:

Find the inverse:

$$3 \cdot 3 = 8 \equiv 1 \pmod 8 \rightarrow 3^{-1} \equiv 3 \pmod 8.$$

Therefore:

$$x \equiv 4 \cdot 3 = 12 \equiv 4 \pmod 8$$
$$x \equiv 4 \pmod 8$$

Now combining using CRT

Since $\gcd(7,8) = 1$, $N = 7 \cdot 8 = 56$

Now let $x = 6 + 7k$, $k \in \mathbb{Z}$

Using the second congruence:

$$6 + 7k \equiv 4 \pmod 8 \longrightarrow 7k \equiv -2 \equiv 6 \pmod 8$$
$$-k \equiv 6 \pmod 8 \rightarrow k \equiv -6 \equiv 2 \pmod 8$$

So $k = 2 + 8t$, $t \in \mathbb{Z}$; $x = 6 + 7(2 + 8t) = \boxed{20 + 56k = x}$

2. $p = 11$, $q = 29$, $n = pq = 319$, $e = 3$

$\varphi(n) = (p-1)(q-1) = 10 \cdot 28 = 280$

Now find $d = e^{-1} \bmod \varphi(n)$:

$$3d \equiv 1 \pmod{280}$$

$$280 = 3 \cdot 93 + 1 \rightarrow 1 = 280 - 3 \cdot 93$$

$$1 \equiv -3 \cdot 93 \rightarrow 3 \cdot (-93) \equiv 1$$

Our inverse of 3 (mod 280) is:

$$-93 \equiv 280 - 93 = \boxed{187 = d}$$

Bob Public Key $P := (3, 319)$; Private Key $S := (187, 319)$

Alice calculates $C := M^e \bmod n = 100^3 \bmod 319$

$$100^2 = 10000 \equiv 10000 - 31 \cdot 319 = 111$$

Then

$$100^3 \equiv 100^2 \cdot 1000 \equiv 111 \cdot 100 = 11000$$

$$11000 - 34 \cdot 319 = \boxed{154 = C}$$

Bob decrypts by calculating $C^d \bmod n$:

$$154^{187} \bmod 319$$ (used calculator)

$$\boxed{121}$$

3. In a graph each vertex can have degree from 0 to n-1:

- Degree 0: isolated vertex
- Degree n-1: vertex connected to all other vertices

This yields n possible degree values: $\{0, 1, 2, \ldots, n-1\}$

By logic, a graph cannot simultaneously contain both degree 0 and n-1, because is some vertex v has degree n-1, then v is connected to all other n-1 vertices in the graph. Specifically, v must be connected to some vertex w of degree 0. But this is a contradiction.

This means the actual possible range of degrees is:

$$\{0, 1, 2, \ldots, n-2\} \vee \{1, 2, \ldots, n-1\}$$

The max is n-1 possible degree values.

Since we have n vertices to apply degrees to and only n-1 possible degrees, at least 2 vertices must have the same degree by Pigeonhole.

4.

4.1) Since G is connected and $2k$ regular, every vertex has degree $2k$. Therefore G has a Eulerian circuit C.

Let m be # edges in G.

We can now construct a subgraph and show its k regular

1. Traverse the Eulerian circuit and alternately color half the edges red and the other half blue.
2. This means we have $\frac{m}{2}$ red and $\frac{m}{2}$ blue edges.
3. Let H be the subgraph of red edges with the same vertex set as the other subgraph.

Now for the Eulerian circuit,

- We must enter and exit v k times since each entry/exit pair accounts for 2 edges at v w/ degree $2k$.
- Each time we enter v on an edge of one color, we exit on an alternate color.
- Therefore, v is incident to k red and k blue edges.

Therefore v has degree k in H, so $\boxed{H \text{ is } k\text{-regular.}}$

4.2) Let G have n vertices. Since G is $2k$ regular:

- Sum of degrees: $2k \cdot n$
- # edges in $G = 2kn/2 = kn$ (kn is odd)

Now H is a k regular subgraph with the same vertex set (n vertices). Therefore:

- Sum of degrees: $k \cdot n$
- # of edges in H: $kn/2 \notin \mathbb{Z}$

This is a contradiction, since the number of edges must be an integer. Therefore, no k regular subgraph H with the same vertex set can exist.

5. G is planar, $v=8$, $e=13$

By Euler's formula,

$$v-e+f=2$$
$$8-13+f=2$$
$$f=7$$

So G has 7 faces.

Assume (for contradiction), G is biparate. Then G contains no odd cycles.

In this planar embedding, each face is bounded by a closed walk, say with L_i being the length of the boundary walk of face i.

Each edge is counted twice when we sum boundary lengths over all faces:

$$\sum_{i=1}^{7} L_i = 2e = 2(13) = 26$$

Since G is a simple graph, $L_i \geq 3$

But if G is biparate, the boundary walk of each face must alternate between the two colors so to return to the same vertex, it must have even length.

Therefore if G is biparate, $L_i \geq 4$.

If $L_i \geq 4$ for all 7 faces:

$$\sum_{i=1}^{7} L_i \geq 7 \times 4 = 28$$

Since $26 < 28$, this is a contradiction and

$\boxed{\text{a planar graph with 8 vertices and 13 edges cannot be 2 colored.}}$